THE LOCALIZATION OF A MORPHISM OF TOPOI ALONG A MORPHISM OF SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the commutativities are reduced to the two companion diagrams and the description of the pullback is obtained by combining the two available descriptions; the closing 'which is formal' concerns an identity of fibre products, a routine verification, and it is preserved verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The localization of a morphism of topoi. Same chapter, Lemma lemma-localize-morphism-topoi. Supplies the morphism between localized topoi whose description is used in the proof.
- Relocalizing a localization of a topos. Same chapter, Lemma lemma-relocalize-topos. Supplies the commutativity of the second square and the description of the pullback of the second localization.

Source reference. Original lemma 04H1: sources/stacks-sites/pinned-original-source.tex, source lines 7904–7904. Navigation pointer invoked at proof line 7904; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04IR: sources/stacks-sites/pinned-original-source.tex, source lines 7951–7951. Navigation pointer invoked at proof line 7951; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be a morphism of topoi. Let $\mathcal{G} \in Sh(\mathcal{D})$, $\mathcal{F} \in Sh(\mathcal{C})$ and $s : \mathcal{F} \to f^{-1}\mathcal{G}$ a morphism of sheaves. There exists a commutative diagram of topoi*

$$\begin{array}{ccc} Sh(\mathcal{C})/\mathcal{F} & \xrightarrow{j_{\mathcal{F}}} & Sh(\mathcal{C}) \\ {\scriptstyle f_s}\downarrow & & \downarrow{\scriptstyle f} \\ Sh(\mathcal{D})/\mathcal{G} & \xrightarrow{j_{\mathcal{G}}} & Sh(\mathcal{D}). \end{array}$$

We have $f_s = f' \circ j_{\mathcal{F}/f^{-1}\mathcal{G}}$ where $f' : Sh(\mathcal{C})/f^{-1}\mathcal{G} \to Sh(\mathcal{D})/\mathcal{F}$ is as in Lemma 04H1 and $j_{\mathcal{F}/f^{-1}\mathcal{G}} : Sh(\mathcal{C})/\mathcal{F} \to Sh(\mathcal{C})/f^{-1}\mathcal{G}$ is as in Lemma 04IR. The functor $(f_s)^{-1}$ is described by the rule

$$(f_s)^{-1}(\mathcal{H} \xrightarrow{\varphi} \mathcal{G}) = (f^{-1}\mathcal{H} \times_{f^{-1}\varphi, f^{-1}\mathcal{G}, s} \mathcal{F} \to \mathcal{F}).$$

Finally, given any morphisms $b : \mathcal{G}' \to \mathcal{G}$, $a : \mathcal{F}' \to \mathcal{F}$ and $s' : \mathcal{F}' \to f^{-1}\mathcal{G}'$ such that

$$\begin{array}{ccc} \mathcal{F}' & \xrightarrow{s'} & f^{-1}\mathcal{G}' \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle f^{-1}b} \\ \mathcal{F} & \xrightarrow{s} & f^{-1}\mathcal{G} \end{array}$$

commutes, then the diagram

$$\begin{array}{ccc} Sh(\mathcal{C})/\mathcal{F}' & \xrightarrow{j_{\mathcal{F}'/\mathcal{F}}} & Sh(\mathcal{C})/\mathcal{F} \\ {\scriptstyle f_{s'}}\downarrow & & \downarrow{\scriptstyle f_s} \\ Sh(\mathcal{D})/\mathcal{G}' & \xrightarrow{j_{\mathcal{G}'/\mathcal{G}}} & Sh(\mathcal{D})/\mathcal{G}. \end{array}$$

commutes.

Proof. The commutativity of the first square follows from the commutativity of the diagram in Lemma 04IR and the commutativity of the diagram in Lemma 04H1. The description of f_s^{-1} follows on combining the descriptions of $(f')^{-1}$ in Lemma 04H1 with the description of $(j_{\mathcal{F}/f^{-1}\mathcal{G}})^{-1}$ in Lemma 04IR. The commutativity of the last square then follows from the equality

$$f^{-1}\mathcal{H} \times_{f^{-1}\mathcal{G}, s} \mathcal{F} \times_{\mathcal{F}} \mathcal{F}' = f^{-1}(\mathcal{H} \times_{\mathcal{G}} \mathcal{G}') \times_{f^{-1}\mathcal{G}', s'} \mathcal{F}'$$

which is formal. □